

SSS 1-3 Research and Design Garden • Practical Activity Cards

Progression: DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

PA-481	Write an Engineering Problem Statement	Complete the investigation using explicit assumptions, systematic measurements, and...
PA-482	Translate Stakeholder Needs	Complete the investigation using explicit assumptions, systematic measurements, and...
PA-483	Define Design Requirements	Define the objective, boundary, variables, constraints, and performance criteria. D...
PA-484	Identify Design Variables	Define the objective, boundary, variables, constraints, and performance criteria. D...
PA-485	Build a Decision Matrix	Define the objective, boundary, variables, constraints, and performance criteria. D...
PA-486	Analyze Design Trade-Offs	Define the objective, boundary, variables, constraints, and performance criteria. D...
PA-487	Develop a Prototype Test Plan	Define the objective, boundary, variables, constraints, and performance criteria. D...
PA-488	Create a Technical Risk Register	Complete the investigation using explicit assumptions, systematic measurements, and...
PA-489	Calculate a Techno-Economic Metric	Use the specified measurements, equation, and assumptions. Show units throughout, c...
PA-490	Conduct an Engineering Design Review	Define the objective, boundary, variables, constraints, and performance criteria. D...